

GNC final project
CubeSat docking through tracking model predictive control

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1 Project idea

This paper, takes inspiration from the article Tracking Model Predictive Control for Docking Manoeuvres of a CubeSat with a Big Spacecraft by Fabrizio Stesina [2] to implement a Tracking Model Predictive Control (TMPC) for a docking manoeuvre. The purpose of the paper is to evaluate the performance of the TMPC in case of a time varying reference trajectory chasing and testing the same controller in case of a thruster failure.

2 Introduction

For the docking between the satellites, the following scenario is considered. A 12 unit CubeSat, chaser, is at the final stage of the docking approach with a satellite, target. The docking is to be considered actuated along the z axis of the body frames. Additional assumptions are introduced:

- The CubeSat is at a range of 50 m from the docking port
- The chaser has a mass of $m_C = 20$ kg while the target has a mass of $m_T = 2000$ kg
- The chaser and target are already aligned and their attitude is controlled and is maintained constant. Inside the Chaser, torque produced by residual thrust are considered to be taken care of through a reaction wheel mechanism.

3 Dynamics and disturbances

For the simulation, three reference frames are employed: the **Earth-Centered Inertial (ECI) frame**, the **local orbital frame**—also referred to as the R–V–H or LVLH frame—and the spacecraft **body frames**.

The absolute motion of both the target and the chaser spacecraft is described in the ECI frame. This inertial frame has its origin at the Earth’s centre of mass, with axes fixed relative to distant stars, making it suitable for the propagation of orbital dynamics under gravitational and perturbative forces.

The local orbital frame (R–V–H/LVLH) is defined with its origin at the centre of mass of the target spacecraft. The axes are constructed such that the X-axis points along the instantaneous orbital velocity vector, the Z-axis is radially aligned from the Earth’s centre of mass to the spacecraft, and the Y-axis completes the right-handed coordinate system, pointing opposite to the orbit’s angular momentum vector. As this frame both translates and rotates along the orbit, it provides a natural representation for relative motion and is therefore well suited for proximity operations and rendezvous problems involving multiple spacecraft.

In contrast, the body frame of each spacecraft is fixed to the spacecraft structure and rotates with it. Body frames are primarily used for attitude representation and control. In scenarios where a specific relative orientation between the target and chaser is of interest—such as during the final docking phase—the respective body frames of the two spacecraft provide the most appropriate reference for defining and enforcing attitude constraints.

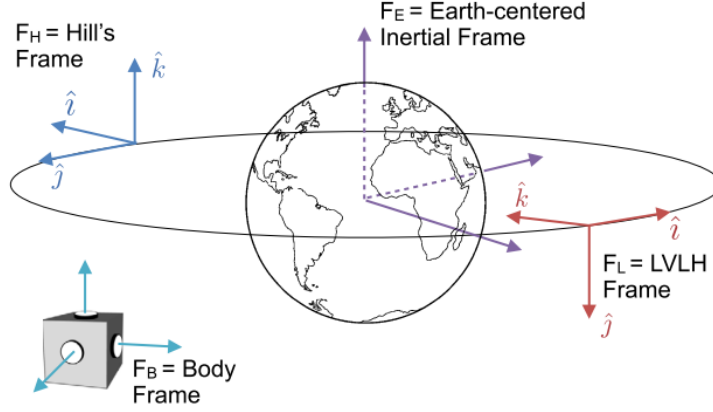


Figure 1: Illustration showing the ECI frame, LVLH frame and the body frame.

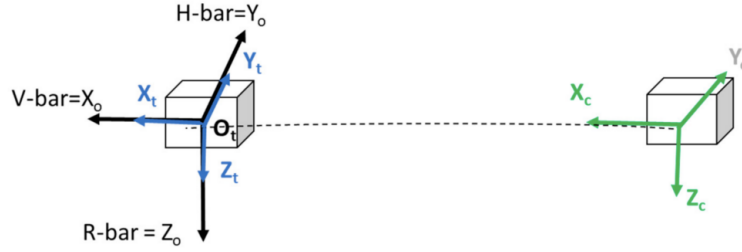


Figure 2: Illustration showing how the R-V-H/LVLH frame is defined for this problem. [2].

The dynamics of the target and chaser spacecraft are simulated independently by propagating their respective equations of motion in the ECI frame. The initial conditions correspond to circular low Earth orbits with mean altitudes of 500 km for the target and 500 km minus 50 m for the chaser, with the relative separation introduced along the radial (z) direction. This formulation enables an accurate representation of the absolute orbital motion while avoiding linearizing assumptions at this stage of the simulation.

Each spacecraft is modelled using a six-dimensional state vector comprising three position components and three velocity components expressed in the inertial frame. The target and chaser state vectors are propagated under identical environmental models, ensuring consistency in the representation of gravitational and perturbative effects. Relative motion quantities used for guidance and control are subsequently derived by transforming the absolute inertial states into the local orbital (LVLH) frame associated with the target spacecraft.

$$\mathbf{X}_s = [\mathbf{r}_s^T, \mathbf{v}_s^T] \in \mathbb{R}, \quad s \in \{T, C\} \quad (1)$$

The simulation and dynamics parameters, variables, and constants used for the dynamics calculations are summarised in the following table. These parameters define the physical properties of the chaser and target spacecraft, their initial absolute and relative states, and the environmental conditions considered in the orbital propagation, including Earth's gravitational field, J2 perturbation, and atmospheric drag at the specified orbital altitude.

Parameter	Value
Chaser mass m_c [kg]	20
Target mass m_t [kg]	2000
Chaser cross-sectional area A_c [m ²]	0.12
Target cross-sectional area A_t [m ²]	10
Chaser initial relative position $r_{t,c,0}$ [m/s]	[0; 0; -50]
Chaser initial relative velocity $v_{t,c,0}$ [m]	[0; 0; 0]
Target initial absolute position r_{t0} [km]	[0; 0; 6871]
Chaser initial absolute position r_{c0} [km]	[0; 0; 6870.950]
Atmospheric density at orbit altitude ρ [kg/m ³]	7.30×10^{-13}
Drag coefficient C_d [-]	2.2
Earth's gravitational parameter μ [m ³ /s ²]	3.986×10^{14}
Earth's J2 coefficient J_2 [-]	1.08263×10^{-3}
Earth's radius R_E [m]	6371×10^3

Table 1: Parameters, variables and constants used.

The dynamics of each spacecraft are governed by Newton's second law, modelling the translational motion as that of a point mass subjected to Earth's central gravitational attraction and additional external accelerations arising from environmental disturbances and control inputs.

$$\dot{\mathbf{r}}_s = \mathbf{V}_s, \quad \dot{\mathbf{V}}_s = \mathbf{a}_{grav,s} + \mathbf{a}_{dist,s} + \mathbf{a}_{ctrl,s} \quad (2)$$

Where the gravitational acceleration around Earth is described as

$$\mathbf{a}_{grav,s} = -\mu \frac{\mathbf{r}_s}{\|\mathbf{r}_s\|^3} \quad (3)$$

The disturbances include J2 disturbances, described as

$$\mathbf{a}_{J2,s} = \begin{bmatrix} k \cdot r_{s,1} \left(\frac{5r_{s,3}^2}{\|\mathbf{r}_s\|^2} - 1 \right) \\ k \cdot r_{s,2} \left(\frac{5r_{s,3}^2}{\|\mathbf{r}_s\|^2} - 1 \right) \\ k \cdot r_{s,3} \left(\frac{5r_{s,3}^2}{\|\mathbf{r}_s\|^2} - 3 \right) \end{bmatrix} \quad (4)$$

where

$$k = \frac{3}{2} \cdot \frac{\mu \cdot J_2 \cdot R_E^2}{\|\mathbf{r}_s\|^5} \quad (5)$$

and atmospheric drag, describes as

$$\mathbf{a}_{drag,s} = -\frac{1}{2} \cdot \frac{C_d \cdot A_s}{m_s \cdot \rho \cdot \|\dot{\mathbf{r}}_s\| \cdot \dot{\mathbf{r}}_s} \quad (6)$$

Note that the atmospheric density ρ is assumed constant and is evaluated at the nominal orbital altitude of 500 km. This simplification is commonly adopted for low Earth orbit analyses to reduce model complexity; however, it neglects variations due to altitude changes, solar activity, and atmospheric conditions. For improved fidelity, the density may instead be modelled as a function of the spacecraft's instantaneous altitude using an exponential atmosphere model or more advanced empirical models.

The control acceleration represents the effect of thrust generated by onboard actuators and is applied exclusively to the chaser spacecraft. The target spacecraft is assumed to be uncontrolled, and

thus no control acceleration is included in its equations of motion. This assumption reflects a passive target scenario and simplifies the relative motion analysis by isolating the control effort to the chaser vehicle.

$$\mathbf{a}_{ctrl,T} = 0, \quad \mathbf{a}_{ctrl,C} = \frac{\mathbf{F}_c}{m_c} \quad (7)$$

Where $\mathbf{F}_c$ is the force vector of the chaser spacecraft, applied by the controller in later stages.

Modelling assumptions and simplifications

- Both spacecraft are modelled as point masses; attitude dynamics and torques are neglected
- Earth is assumed spherical for the central gravity term; oblateness is represented only via the J2 disturbance
- Atmospheric rotation and winds is neglected, the atmospheric density is assumed constant
- Higher-order gravitational harmonics, solar radiation pressure and third-body effects are omitted

With a constant thrust of 1 N applied to the chaser spacecraft, the coupled translational dynamics of the chaser–target system are simulated by numerically propagating their respective equations of motion under the defined environmental models. The resulting absolute and relative motion profiles over one orbital period are presented in Figure 3, illustrating the effect of sustained low-magnitude control input on the evolution of the spacecraft separation and orbital trajectories.

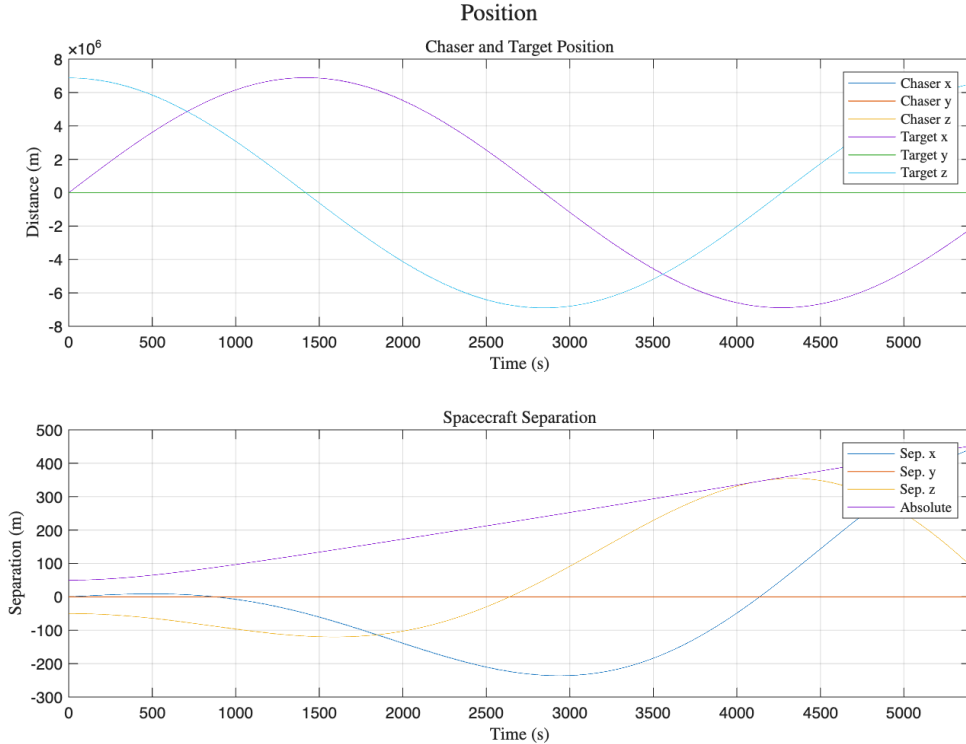


Figure 3: Actual position and relative position of the two spacecraft during one orbital period.

4 Tracking model predictive control

The control is achieved through a TMPC which takes into account the control horizon of the immediate future. To minimize processing cost and time the future horizon is limited to $H_p = 6$ steps in the

future, which means for a simulation running with a fixed stepsize of 0.5 s the controller is predicting 3 s into the future.

The relative state coming from the dynamics blocks is fed to the controller after introducing a $\pm 5\%$ error to simulate sensor errors. The controller at each steps reads the relative state and computes the control command needed by minimizing the cost function derived as follows.

Linearised dynamics are introduced through the Hill-Clohessy-Wiltshire according to the close proximity operations assumption.

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2\Omega \\ 0 & -\Omega^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3\Omega^2 & -2\Omega & 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1/m_C & 0 & 0 \\ 0 & 1/m_C & 0 \\ 0 & 0 & 1/m_C \end{pmatrix} \quad (8)$$

This dynamic model has to be converted from continuous time to discrete time in order to predict the correct future state given the sample time of the controller. The transition from continuous to discrete can be easily achieved through MATLAB function `c2d`.

Future dynamics can now be predicted using the new discrete time matrices through the following equations:

$$\mathcal{A} = (A \quad A^2 \quad \dots \quad A^{H_p})^T \quad \mathcal{B} = \begin{pmatrix} B & 0 & \dots & 0 & 0 \\ AB & B & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots \\ A^{H_p-2}B & A^{H_p-1}B & \dots & B & 0 \\ A^{H_p-1}B & A^{H_p-2}B & \dots & A^2B & AB \end{pmatrix} \quad (9)$$

The dynamics equation is now written using the augmented states and commands:

$$\mathcal{X}(k) = \mathcal{A}x(k|k) + \mathcal{B}u(k) \quad (10)$$

where:

$$\begin{aligned} \mathcal{X}(k) &= [x(k|k) \quad x(k+1|k) \quad \dots \quad x(k+H_p-1|k)] \\ \mathcal{U}(k) &= [u(k|k) \quad u(k+1|k) \quad \dots \quad u(k+H_p-1|k)] \end{aligned} \quad (11)$$

A reference trajectory can be introduced to take into account the initial error with respect to the desired approach along z axis. This is achieved by imposing the following trajectory for the future states at each time step of the controller:

$$\begin{aligned} r(k+i)|_{x,y} &= x(k|k)|_{x,y} e^{\frac{-t_i}{\tau}} \\ r(k+i)|_z &= x(k|k) + v_{z,nom} t_i \\ r(k+i)|_{v_x,v_y} &= -(r(k+i)|_{x,y} - x(k|k)|_{x,y})/t_i \\ r(k+i)|_{v_z} &= v_{z,nom} \end{aligned} \quad \text{for } i = 0, 1, \dots, H_p - 1 \quad (12)$$

This simple trajectory takes into account the initial position error and slowly brings the chaser onto a linear trajectory along z axis. It is achieved through an exponentially decaying error, through the Glide slope algorithm [3] and the reference velocity is simply calculated as the error between reference and the current state divided by the time. For this purpose the time constant τ is set to 30 s for best performance and nominal vertical velocity $v_{z,nom}$ is set to 0.12 m/s until the chaser reaches 6 m from the target at which point the speed is set to zero so at the target encounter the velocity is brought to

the desired values under 0.02 m/s.

At this point the optimization criterion is written by taking into account also the reference trajectory:

$$J(U(k), x(k|k)) = \sum_{i=0}^{H_p-1} (x(k+i|k) - r(k+i))^T Q (x(k+i|k) - r(k+i)) + u(k+i|k)^T R u(k+i|k) \quad (13)$$

Considering the augmented states and augmented weighting matrices Q and R defined as:

$$\mathcal{Q} = \begin{pmatrix} Q & .. & 0 \\ ... & Q & .. \\ 0 & .. & Q \end{pmatrix} \quad \mathcal{R} = \begin{pmatrix} R & ... & 0 \\ .. & R & ... \\ 0 & ... & R \end{pmatrix} \quad (14)$$

The reference trajectory is collected into a vector R defined as:

$$R = [r(k) \quad r(k+1) \quad ... \quad r(k+H_p-1)] \quad (15)$$

The function to minimize, after substituting eq 10 into eq 13:

$$J(U(k), x(k|k)) = (\mathcal{A}x(k|k) + \mathcal{B}U(k) - R(k))^T \mathcal{Q} (\mathcal{A}x(k|k) + \mathcal{B}U(k) - R(k)) + U(k)^T \mathcal{R} U(k) \quad (16)$$

After some simplifications the optimization can be seen in a quadratic programming formulation described by the following cost function:

$$J(U(k), x(k|k)) = U(k)^T H_{opt} U(k) + \begin{pmatrix} x(k|k) \\ R(k) \end{pmatrix}^T F_{opt} U(k) + \bar{F}(x(k|k), R(k))$$

with

$$H_{opt} = \mathcal{B}^T \mathcal{Q} \mathcal{B}$$

$$F_{opt} = 2 \begin{pmatrix} \mathcal{A}^T \mathcal{Q} \mathcal{B} \\ \mathcal{Q} \mathcal{B} \end{pmatrix} \quad (17)$$

The function $\bar{F}$ is not dependant on U thus can be ignored in the minimization. **MATLAB** function **quadprog** is used with algorithm “active-set” and constraints on the commanded thrust to have the same limitations as the onboard thrusters maximum performance.

$$-F_{max} \leq U \leq F_{max} \quad \text{with } F_{max} = 0.035N \quad (18)$$

As authors of the original paper declare the controller performance heavily depends on the algorithm used for the quadratic programming solution. The solution is filtered and only the command for the current iteration is selected and sent to the thrusters. If the algorithm failed to find a solution the requested thrust is set to zero for fail proof the controller.

5 Thruster modelling and thruster failure

The chaser spacecraft is assumed to be equipped with a set of 8 cold-gas thrusters capable of generating translational forces along the three axes of the local orbital frame. Each thruster is oriented with a 45° angle as shown in Figure 4. The maximum thrust in one direction is $F_{max} = 0.035N$ as already stated before, while the maximum thrust per single thruster is $F_{max_s} = 12.38 \cdot 10^{-3}N$. To account for the contribution of each thruster along each direction the matrix M is defined as

$$M = \begin{pmatrix} \cos(\alpha) & \cos(\alpha) & -\cos(\alpha) & -\cos(\alpha) & \cos(\alpha) & \cos(\alpha) & -\cos(\alpha) & -\cos(\alpha) \\ \sin(\alpha) & -\sin(\alpha) & \sin(\alpha) & -\sin(\alpha) & \sin(\alpha) & -\sin(\alpha) & \sin(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \sin(\alpha) & \sin(\alpha) & \sin(\alpha) & -\sin(\alpha) & -\sin(\alpha) & -\sin(\alpha) & -\sin(\alpha) \end{pmatrix} \quad (19)$$

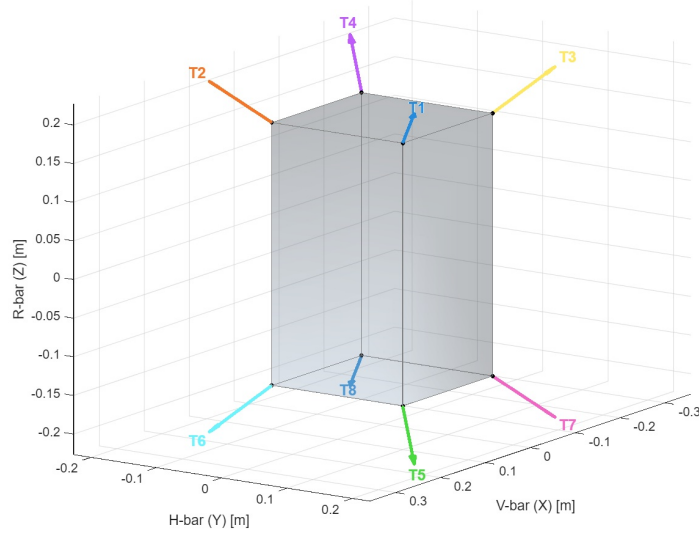


Figure 4: Thruster configuration

where $\alpha = 45^\circ$.

The thruster selection logic implemented is an algorithm used to determine the optimal distribution of forces among the eight thrusters, in order to achieve the desired force vector (F_x, F_y, F_z) . The approach is based on solving a quadratic optimization problem, where the sum of the squares of the individual thruster forces is minimized, subject to the physical constraints of each actuator. Mathematically, the problem is formulated as

$$M\mathbf{T} = \mathbf{F} \quad (20)$$

where M is the thruster contribution matrix and $\mathbf{T} = [T_1, \dots, T_8]^\top$ is the vector of thruster forces. The optimization ensures that the resulting force vector $\mathbf{F}$ is achieved while respecting the saturation limits of each thruster. After solving the quadratic problem, the thrust commands are clipped to remain within the allowed bounds.

Once the optimal thruster forces T_i are computed and saturated within their physical limits, each force command is converted into a corresponding Pulse Width Modulation (PWM) signal to drive the thruster actuators. This step closes the control loop, allowing the vehicle to generate the required force vector $\mathbf{F}$ in real-time.

In order to model the failure of a single thruster, a vector of availability flags is introduced:

$$T_{\text{available}} = [1 \quad 1 \quad \dots \quad 0 \quad \dots \quad 1]^\top \quad (21)$$

where a value of 1 indicates that the thruster is operational, and 0 represents a failed thruster. For example, setting $T_{\text{available},i} = 0$ simulates the failure of the thruster i .

The actual PWM commands sent to the thrusters are then adjusted by element-wise multiplication with the availability vector:

$$T_{\text{actual}} = T_{\text{requested}} \odot T_{\text{available}} \quad (22)$$

where $T_{\text{requested}}$ are the PWM signals computed from the desired thrusts. This approach effectively disables the failed thruster while leaving the remaining thrusters unaffected, allowing the system to continue operating under partial actuator failure.

Finally, the actual force vector generated by the thrusters is obtained by multiplying the thruster commands with the contribution matrix:

$$\mathbf{F}_{\text{actual}} = \mathbf{M} \mathbf{T}_{\text{actual}} \quad (23)$$

where $\mathbf{T}_{\text{actual}}$ are the PWM adjusted thrusts that account for actuator saturation and potential failures. This operation maps the individual thruster outputs back into the resulting force in the spacecraft frame, completing the control loop from the desired force vector $\mathbf{F}$ to the actual force $\mathbf{F}_{\text{actual}}$ applied by the chaser spacecraft. As shown in Figure 5, the actual force generated by the thrusters closely follows the force requested by the TMPC controller along all three axes. Small discrepancies between the requested and actual forces are mainly due to actuator saturation and PWM discretization effects. Overall, the force tracking performance is satisfactory, demonstrating that the thruster allocation logic combined with the PWM conversion effectively reproduces the commanded control forces under nominal operating conditions.

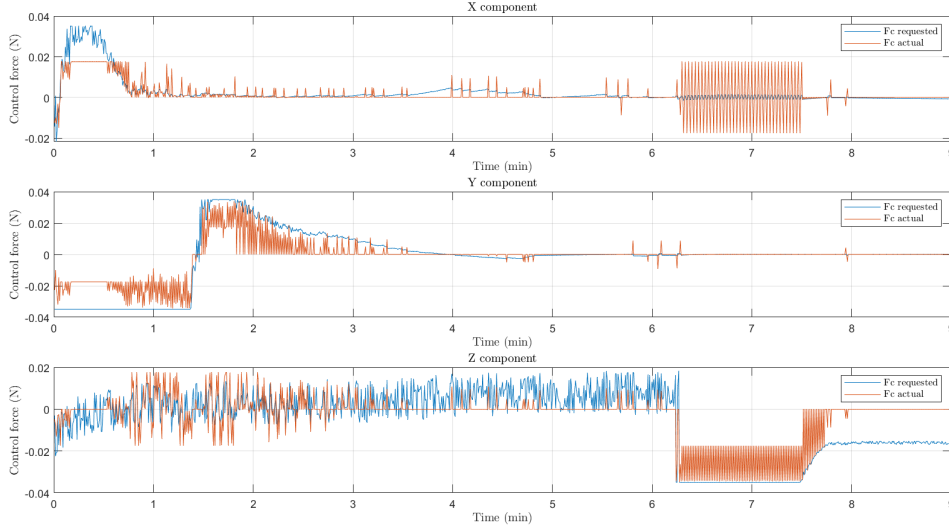


Figure 5: Control force requested from TMPC and actual control force from the thrusters

6 Controller performance

The initial relative state is set as follows:

$$\begin{aligned} r_{rel,0} &= [0, 0, -50]m \\ v_{rel,0} &= [0, 0, 0]m/s \end{aligned} \quad (24)$$

The position has an uncertainty of ± 2 m on x and y axis, while the control is assumed to start as soon as the chaser is at 50 m from target. The velocity can have up to ± 0.05 m/s of initial residual relative speed while on z the velocity can have uncertainties up to 0.2 m/s.

After few Monte Carlo simulations with different Q and R matrices values, iterated around the values proposed in the paper, the following matrices were found to achieve the best performance to cost ratio

scenario.

$$Q = \begin{pmatrix} 300 & 0 & 0 & 0 & 0 & 0 \\ 0 & 300 & 0 & 0 & 0 & 0 \\ 0 & 0 & 28 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 200 \end{pmatrix} \quad R = \begin{pmatrix} 10 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 10 \end{pmatrix} \quad (25)$$

6.1 Monte Carlo simulation with no thruster failure

A Monte Carlo (MC) simulation consisting of 100 random initial conditions within the given uncertainty is carried out. The initial values for the test campaign are shown in table 2. The resulting trajectories are displayed in figure 6. The expected performance according to the reference paper are found in table 3.

Parameter	Value
Initial relative position [m]	$[\pm 2, \pm 2, 50]$
Initial relative velocity [m/s]	$[\pm 0.05, \pm 0.05, 0.2]$

Table 2: Monte Carlo Analysis parameters

Parameter	Value
Approach velocity along z axis [m/s]	< 0.02
Lateral alignment [m]	< 0.01
Lateral velocity [m/s]	< 0.01
Manoeuvre time [min]	< 10

Table 3: Expected performance

The time requirement comes from the typical observation window of the manoeuvre from a ground station.

The resulting mean performance is grouped in table 4.

Performance parameter	Mean Value
Alignment error x axis [m]	5.3×10^{-4}
Alignment error y axis [m]	7.5×10^{-6}
Approach velocity x axis [m/s]	2.7×10^{-5}
Approach velocity y axis [m/s]	2.1×10^{-6}
Approach velocity z axis [m/s]	8.9×10^{-3}
Time to approach [s]	559
Total thrust [N]	15

Table 4: MC results

6.2 Results discussion

The results show that the TMPC control is able to carry out the manoeuvre keeping the chaser within the safety cone of aperture angle $\delta = 7.5^\circ$ around z axis of the target spacecraft. This characteristic is achieved without imposing any constraint on the quadratic programming solver, rather by intrinsic construction of the reference trajectory. This choice made the control effort heavier with respect to

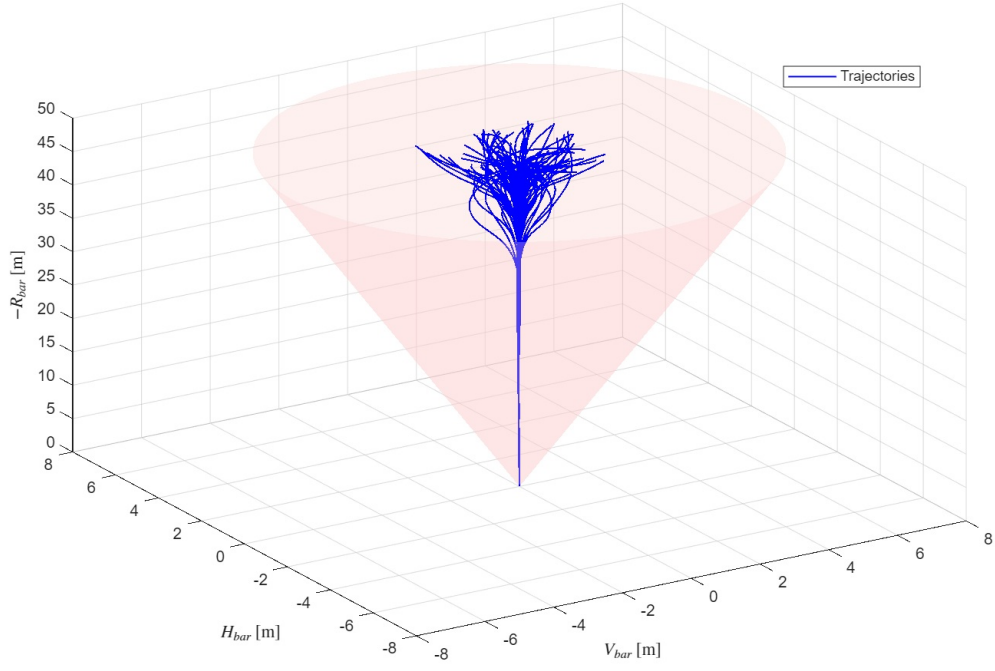


Figure 6: Trajectories and safety cone Monte Carlo simulation with no thruster fail

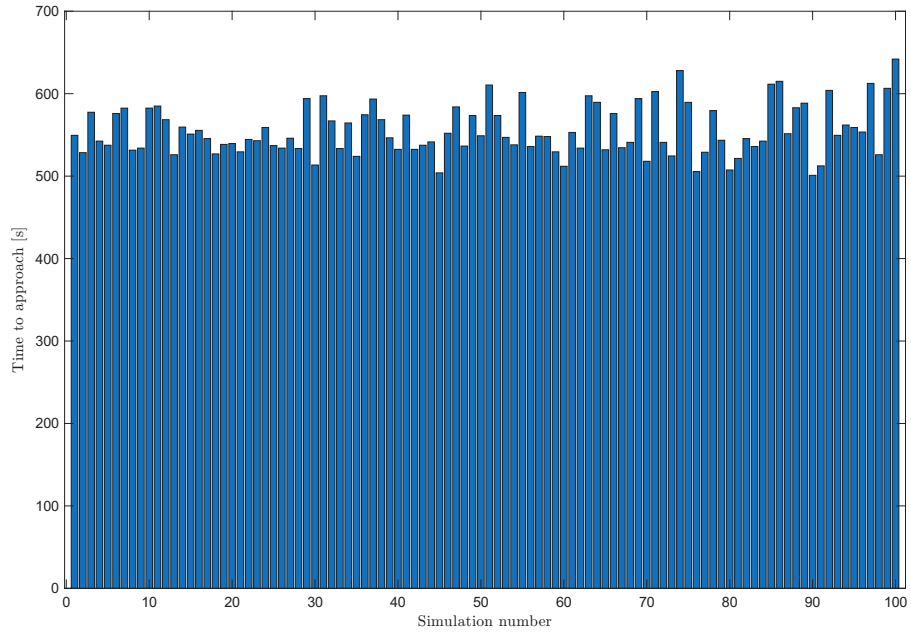


Figure 7: Manoeuvre Time

the reference paper’s results. The manoeuvring time for the MC campaign is found in figure 7. The docking is completed in around 10 minutes in all expected scenarios.

6.3 Monte Carlo simulation with one thruster failure

A Monte Carlo analysis similar to the previous with the addition of the thruster failure is carried out. This will be pushing the TMPC controller to its limit, no feedback is sent to the controller to let it know that the demanded thrust cannot be achieved, so the controller is fighting against it’s wrong prediction at each time. Same initial values as in table 2 are used. While one of the 4 thrusters in the face opposite to the docking face is turned randomly off. These choice is adopted with the assumption that no thrust fail is expected in the most critical face and direction that has to dock and slow down the CubeSat. Resulting trajectories are found in figure 8.

6.4 Results discussion

It is possible to visibly appreciate the difficulty to get to the straight path with respect to the scenario with no failure. The resulting performance index in terms of mean values are grouped in table 5.

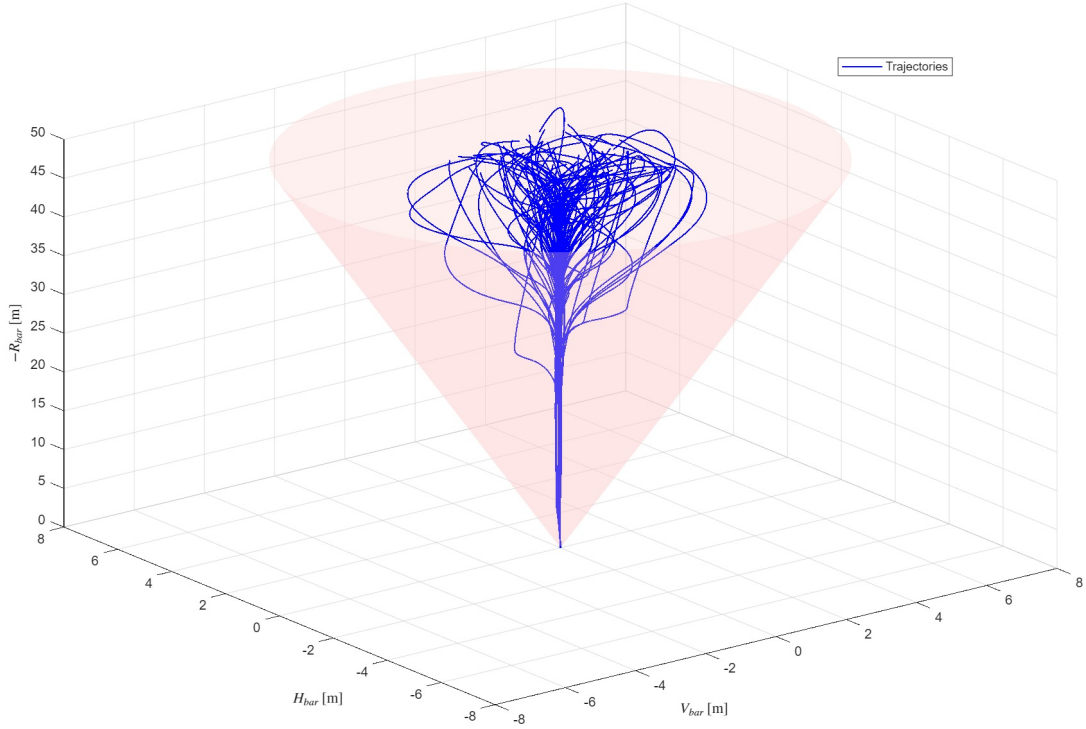


Figure 8: Trajectories and safety cone Monte Carlo simulation with one thruster fail

Despite the initial difficulty to follow the straight path the chaser is still able to achieve the desired docking performance. Note that no thrust misalignment is considered, thus even if the performance in terms of alignment errors seems better they are justified by the longer manoeuvre time resulting from the initial path loss and one must consider that in real life scenario the thruster loss will induce an undesired torque that reaction wheels have to dump continuously. Indeed if the target velocity along z

Performance parameter	Mean Value
Alignment error x axis [m]	1.9×10^{-4}
Alignment error y axis [m]	1.2×10^{-4}
Approach velocity x axis [m/s]	1.5×10^{-5}
Approach velocity y axis [m/s]	1.0×10^{-6}
Approach velocity z axis [m/s]	8.9×10^{-3}
Time to approach [s]	589
Total thrust [N]	15

Table 5: MC results with one thruster failure

axis is reduced in the reference trajectory the manoeuvre will take longer but yields better alignment performance. The time required for the manoeuvre with one thruster fail can be seen in figure 9.

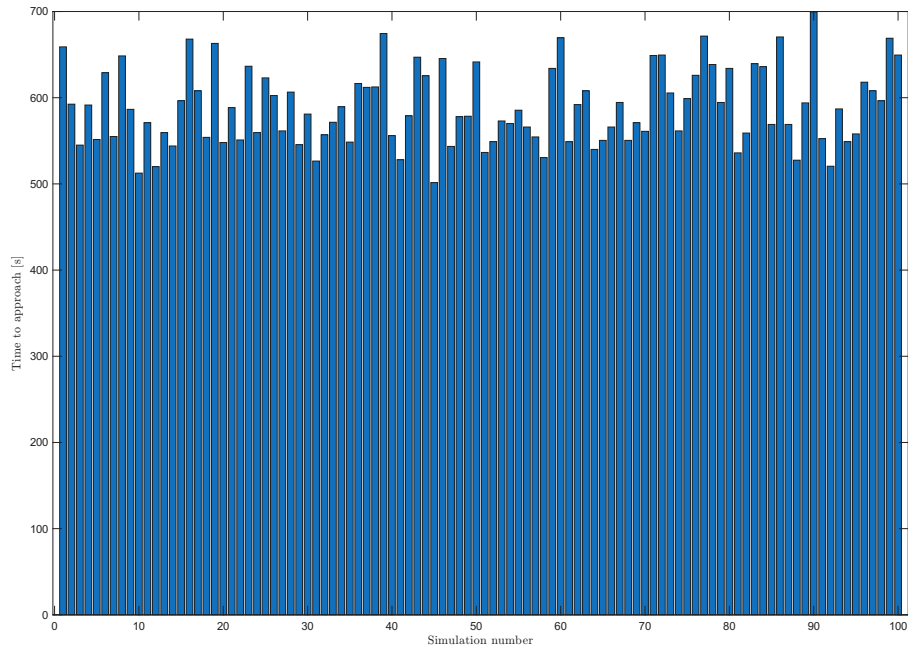


Figure 9: Manoeuvre Time

7 Conclusions

Results from the MC campaigns validates the performance of the TMPC controller for any expected scenario. The desired smooth docking within the safety cone is achieved. Future work could benefit from addition of chaser kinematics and controlling simultaneously position and orientation through a TMPC.

Acronyms

ECI Earth-Centered Inertial. 3, 4

LVLH Local Vertical Local Horizontal. 3, 4

MC Monte Carlo. 11, 13, 14

PWM Pulse Width Modulation. 9, 10

TMPC Tracking Model Predictive Control. 3, 6, 10, 11, 13, 14

References

- [1] Luleå University of Technology. *R7030R: Spacecraft Guidance, Navigation and Control - Course Material*. Lecture notes and course documentation. Kiruna, Sweden, 2025.
- [2] Fabrizio Stesina. “Tracking Model Predictive Control for Docking Maneuvers of a CubeSat with a Big Spacecraft”. In: *Aerospace* 8.8 (2021), p. 197. DOI: 10.3390/aerospace8080197. URL: <https://www.mdpi.com/2226-4310/8/8/197>.
- [3] Hari B. Hablani, Michael L. Tapper, and David J. Dana-Bashian. “Guidance and Control for Autonomous Proximity Operations and Docking with a Target Spacecraft”. In: *Journal of Guidance, Control, and Dynamics* 25.3 (May 2002), pp. 553–562. DOI: 10.2514/2.4916. URL: <https://doi.org/10.2514/2.4916>.